

An overview of our chamber problem

Two errors, not one: the blade tones are bent by up to 8 dB below 1 kHz, and under them sits a broadband error of ±1.3 dB (2.7 dB worst) that no band is exempt from. The arc is not qualified to a standard anywhere yet. For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height in the propeller plane, so a perfect setup reads the same level everywhere: a circle. Twelve runs (prop7–18, 1–2 Sept 2026, PWM 1900); what is left after each microphone's own offset is removed is the room.

2026-09-08 · rev 2
data: laptop backup of 2 Sept
scripts: arc_error_map.py, arc_validation_diagnostics.py
full guide: docs/arc-validation-guide.html
remedies report: docs/arc-validation-remedies.pdf

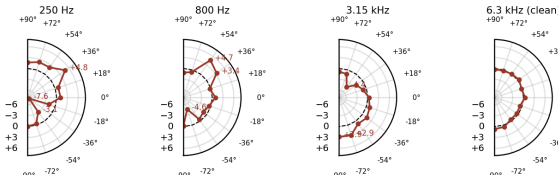


Fig. 1 · room error P(position), dB from a circle, mic offsets removed · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 spot (ids in the table below)

What the room does (right panel of Fig. 2)

How to read the map. Row = spot on the arc, column = frequency band. White: the spot reads what a circle predicts. Red: too loud. Blue: too quiet. Each microphone's own error is already taken out (left panel), so the colour is the room.

Deep blue at –72° and red at +36°/+54°, 500-1000 Hz. A sound wave that bounces off a nearby surface arrives a little later than the direct wave. If the extra path is half a wavelength the two cancel and the spot goes blue; if it is a whole wavelength they add and the spot goes red. The extra path depends on the spot, so neighbours can be opposite colours and one spot can flip colour from band to band. Wavelength here is 35–70 cm, so the surface is within about a metre.

The 200-250 Hz column is the blade tone, and it is not one nearby surface. At BPF 216 Hz the tone polar is round to ±2 dB; at 238 Hz –36° reads –8 dB; at 258 Hz the hole moves to –72° and the peak to +54°, while the broadband floor beside the tone stays round to ±1 dB. A surface within a metre cannot change the pattern that fast; narrow room modes sampled by a line spectrum can, and so can a propeller whose inflow is not axisymmetric. A loudspeaker at the hub decides which (remedies report, §03, §10).

Blue on one side, red on the other: the surface is off-centre. A surface symmetric around the propeller (the floor under a flat arc, say) gives every microphone the same extra path, so a whole row would be one colour. Here the colour changes around the arc, so the microphones are at different distances from the surface: it sits to one side. The colours do not say which side. Probing with the tone at three speeds: the +36°/+54° reflection has an extra path of 1.5 m and a hard surface's amplitude, the –72° one 0.7 m (remedies report, Fig. 3); the unrecorded arc radius is what turns those into places.

Taking the tones out removes the big errors, not all of them. A wide band averages a reflection out, a tone does not, which is why the tones carry the big errors. With every tone notched the polar is round to ±1.3 dB in 92 % of cells, worst 2.7 dB — but 99 of those 209 cells sit more than three standard errors from zero, so the rest is a real pattern, not scatter, and it is no smaller above 3 kHz than below (2.7 dB at +72°/2 kHz, 2.6 at –72°/397 Hz, 2.6 at +54°/5 kHz). The tones need the loudspeaker check or the speed ladder; the broadband floor of ±1–3 dB is there in every band (report §00, §03).

Fig. 2 · Microphone error map (left) and room error map (right), same fit, same scale

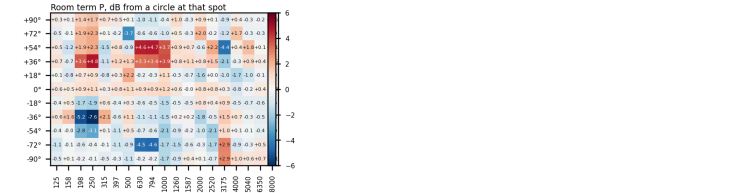
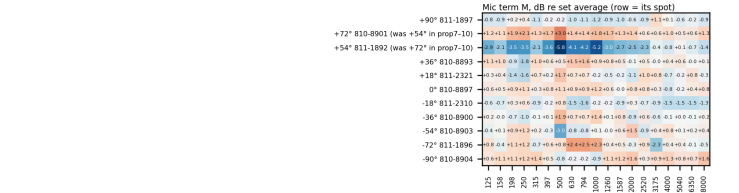


Fig. 2 · one fit over keys dp1-baseline-horizontal-prop7...prop18, PWM 1900, prop11–14 mirrored · rows aligned: mic row i sat at spot row i · 0° mic never moved, so its M and the 0° P are known only as a sum · third-octave centre, Hz

What to look for in the chamber

Three suspects, one per band in Fig. 2. Each is a size and a place, read off the map: a surface shows up once the wavelength is shorter than it (Rajmane and Baumann, DAGA 2016, Table 1: a 35 cm plate from 1 kHz, a 10 cm plate from 3.15 kHz).

- 200-250 Hz, blue at –36°.** Something about 1.5 m across, off to one side of the rig, with a bounced path 0.7 m longer than the direct one at –36°: a wall, a table top or the corner within about half a metre of that microphone.
- 630-1000 Hz, blue at –72°.** Something about 0.4 m across near the –90° end of the arc on the corner side: the stand base, a bar of the arc frame, a cable tray.
- 3150 Hz, scattered red and blue.** Something about 0.1 m across at the microphones themselves: clamps, mounts, the cable loop.

How the numbers were made (not a sum, not an average)

Every level in the twelve runs (11 microphones × 19 third-octave bands × 12 runs) is written as three parts added together: **run gain + position error + microphone offset**. The three parts are found by least squares over all 12 × 11 levels in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **Microphone offset** takes up each serial's own error after calibration. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

The turned-around runs are handled before the fit: for prop11–14 the label +e is entered as physical –e, so “–72°” always means the same spot in the room. Microphone offsets and position errors can be told apart only where one spot was covered by different microphones (the swap after prop10, the four turned runs). Without that, a low microphone and a low spot would be indistinguishable. Spectra: Welch 4096-point Hann, UMIK-2 calibration as the app applies it, third-octave sums 125 Hz–8 kHz. Data: the last PWM-1900 capture in each key 2004_bin_unset_dp1-baseline-horizontal-prop7...prop18, 11 mics × 2.0 s (prop1–6 and the 31 Aug keys match neither orientation and are left out); ids in docs/analysis/arc-validation-data.json. Code: scripts/arc_error_map.py, function fit.

Do the numbers add up?

All numbers re-checked against the source data 2026-09-11, 40 of 40 (AUDIT-2026-09-11.md). Refitting reproduces the maps to 0.01 dB. 132 cells per band are explained by 34 unknowns; 87% of them are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells (+11 dB at –36°/250 Hz; +6 to +7 dB at 0°/2–4 kHz) come from prop7, which spun at BPF 164 Hz against 238 Hz elsewhere; leaving it out moves the maps by under 1 dB. **Limit, found 2026-09-11:** a microphone separates a bad spot from a bad capsule only where it sat in both, and ours visited five disconnected clusters (±90; –72/–54/+54/+72; ±36; ±18; 0). Between clusters the two terms cannot be separated and the solver settles it by minimum norm. Contrasts inside a cluster hold, including the pair below; each spot against the arc mean carries an unmeasured offset of about a decibel, until the microphones are calibrated against each other (report, Problems 4 and 6).

Worked example · –72° spot at 800 Hz, two mics, two orientations

RUN	MIC THERE	MEASURED	GAIN	P ROOM	M MIC	MODEL	LEFT
prop18	811-1896	59.1	62.3	–4.6	+2.5	60.2	–1.1
prop12 (180°)	810-8901	58.6	62.2	–4.6	+1.4	59.0	–0.4

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P + 4.7 + M + 1.4; prop18 with 811-1892 reads +3.1 = +4.7 – 4.2. Two mics, one room value.